MSA102 - Assignment 2

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Question 1a) The data can be visualized as follows.

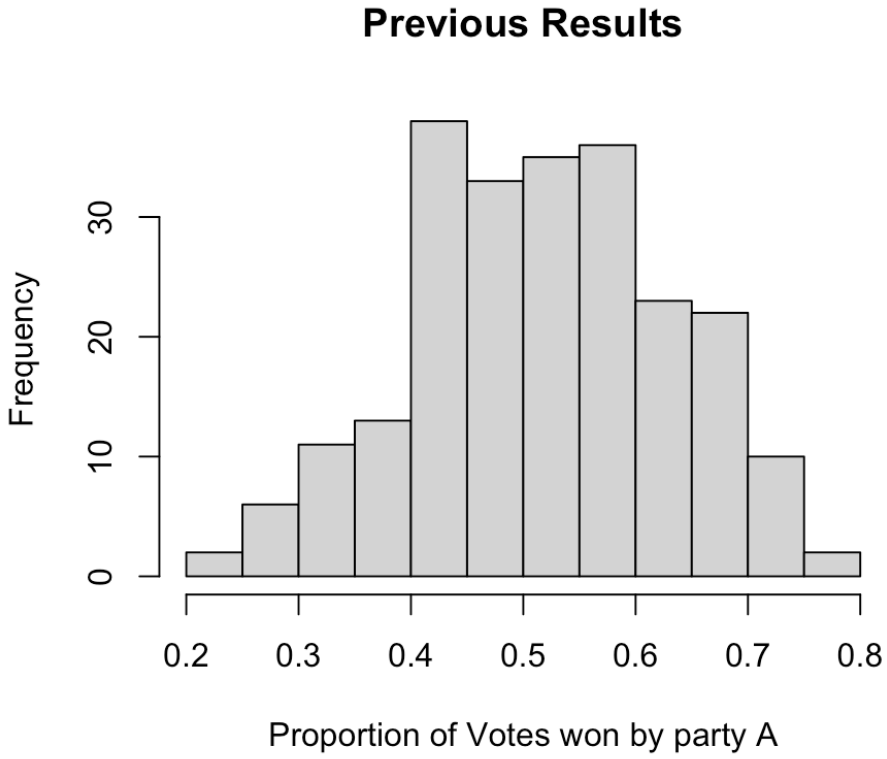


Figure 1: Previous Results

Party A won 128 out of 231 of the seats in the previous election - a comfortable majority.

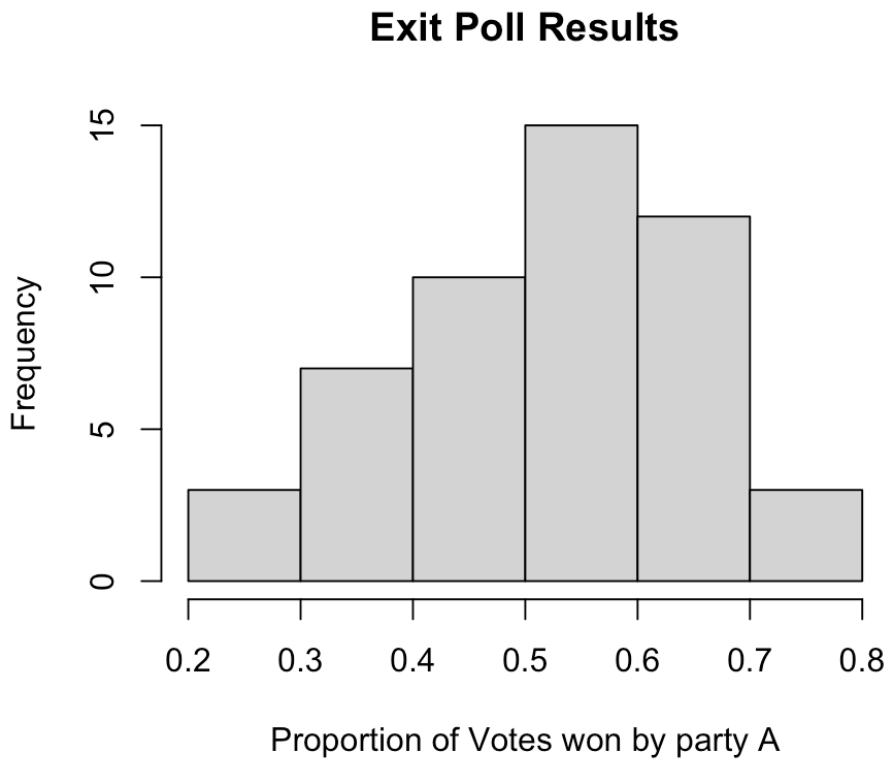


Figure 2: Exit Poll Results

Party A won 30 out of 50 districts selected for exit polling, suggesting a good performance.

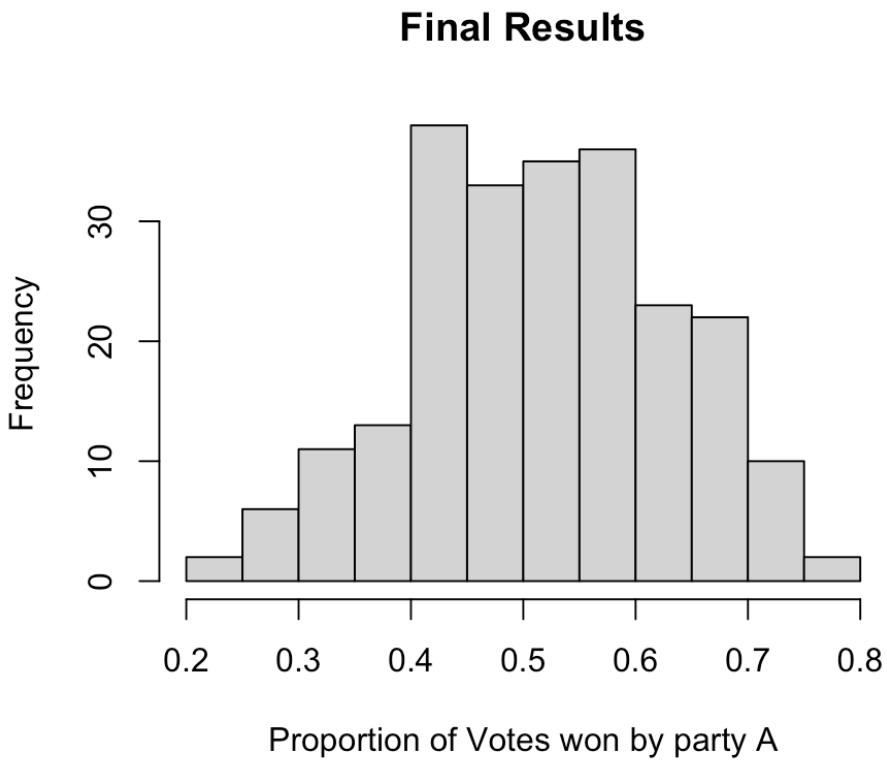


Figure 3: Final Results

Party A won 121 out of 231 seats this election - worse than the previous one, however still 5 more seats above the majority threshold.

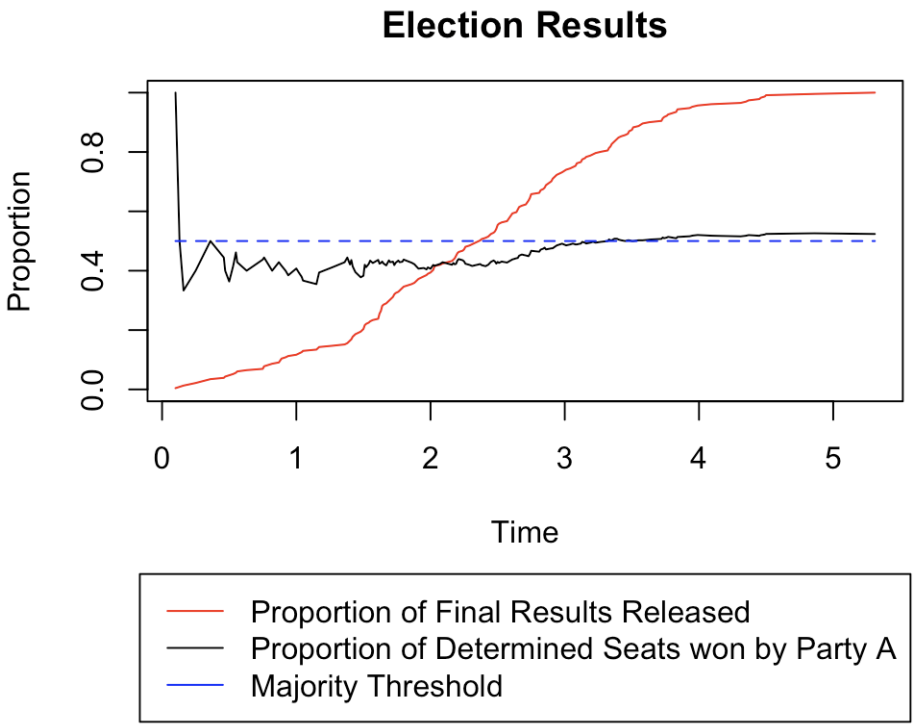


Figure 4: Time Graph of Results

The results come in over the course of 4 hours. It seems that party A is "losing" the election given only data on the final results until roughly 3.5 hours into the election, when it finally won more than half of the seats.

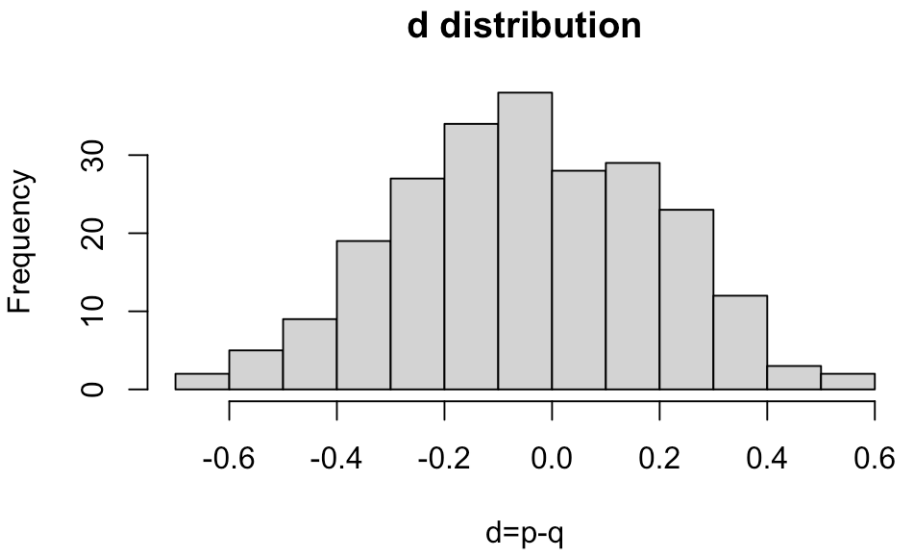


Figure 5: Post-Hoc Distribution of $d = p - q$

Checking the distribution of $d = p - q$ using post-hoc results after the election, we can see that the results roughly do follow a bell shaped curve. The μ and γ values were calculated as $\mu = -0.0445$ (suggesting slightly worse performance this election by part A) and $\gamma = 2.88$ respectively.

The Bayesian network for the described situation can be visualized below.

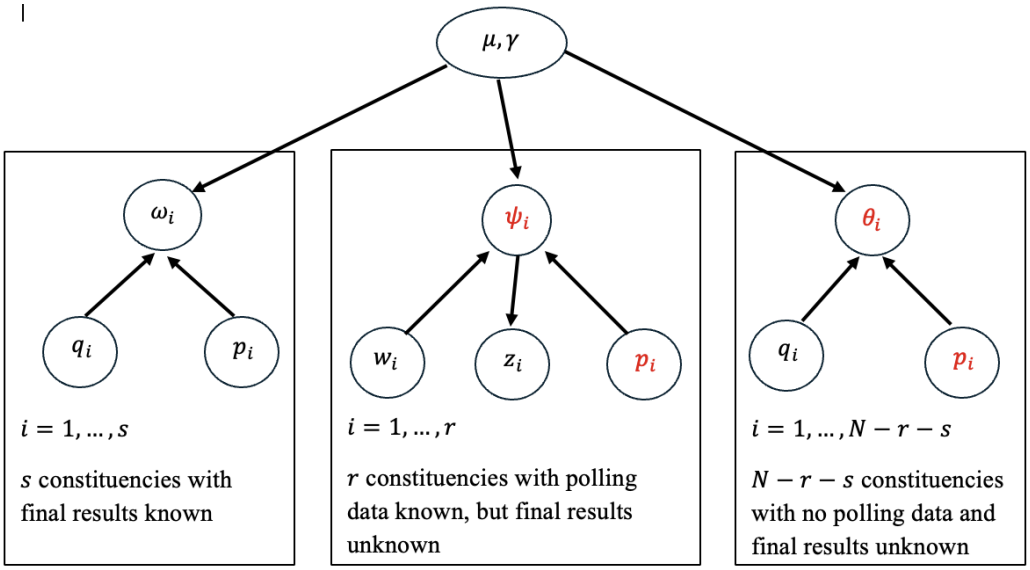


Figure 6: Bayesian network for voting situation

The three d_i distributions represented by ω_i, ψ_i and θ_i are the difference of the log-odds of the proportion of votes that party A received (p_i) vs. the log-odds of the proportion of votes that party A received in the previous election, (q_i and w_i). The nodes in red indicate information that is not yet observed. As all previous elections are known, we can further simplify the Bayesian network as follows:

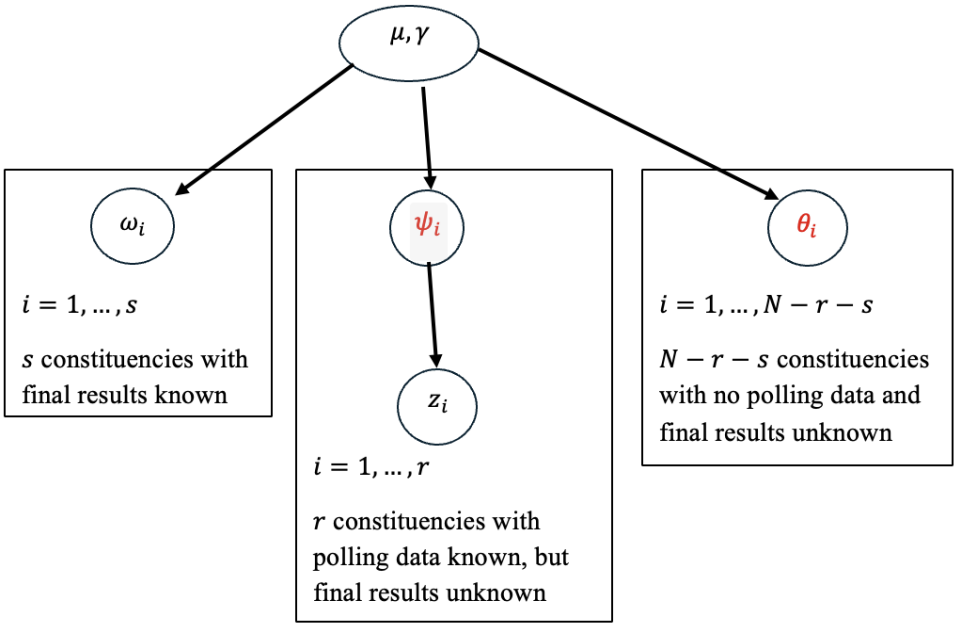


Figure 7: Simplified Bayesian network for voting situation

Question 1b)

$$\begin{aligned}
\pi(\theta, \psi|\omega, z) &= \iint \pi(\theta, \psi, \mu, \gamma|\omega, z) d\mu d\gamma && \text{Marginalizing over } \mu, \gamma \\
&= \iint \pi(\theta|\mu, \gamma, \psi, \omega, z) \cdot \pi(\mu, \gamma, \psi|\omega, z) d\mu d\gamma && \text{Bayes' Theorem} \\
&= \iint \pi(\theta|\mu, \gamma) \cdot \pi(\mu, \gamma, \psi|\omega, z) d\mu d\gamma && \text{All information from } \psi, \omega, z \text{ already included in } (\mu, \gamma), \text{ the Markov Boundary of } \theta
\end{aligned}$$

Question 1c) We can find the negative logarithm of the joint density of all variables first:

$$
\begin{aligned}
&-\log[\pi(\mu,\gamma,\omega,\psi,z,\theta)] \\
&\quad = -\log[\pi(\mu)\cdot\pi(\gamma)\cdot\pi(\omega|\mu,\gamma)\cdot\pi(\psi|\mu,\gamma)\cdot\pi(\theta|\mu,\gamma)\cdot\pi(z|\psi)] \\
&\quad = -\log\Bigg[\prod_{i=1}^{s} Normal(\omega_i;\mu,e^{-\gamma})\prod_{i=1}^{r} Normal(\psi_i;\mu,e^{-\gamma})\prod_{i=1}^{N-r-s} Normal(\theta_i;\mu,e^{-\gamma}) \\
&\qquad \cdot\prod_{i=1}^{r} Binom(z_i;100,\frac{e^{\psi_i+w_i}}{1+e^{\psi_i+w_i}})\Bigg] \\
&\quad = -\sum_{i=1}^{s}\log[Normal(\omega_i;\mu,e^{-\gamma})] - \sum_{i=1}^{r}\log[Normal(\psi_i;\mu,e^{-\gamma})] - \sum_{i=1}^{N-r-s}\log[Normal(\theta_i;\mu,e^{-\gamma})] \\
&\qquad -\sum_{i=1}^{r}\log\left[Binom(z_i;100,\frac{e^{\psi_i+w_i}}{1+e^{\psi_i+w_i}})\right] \\
&\quad = C - \sum_{i=1}^{s}\left[\frac{\gamma}{2}-\frac{1}{2e^{-\gamma}}(\omega_i-\mu)^2\right] - \sum_{i=1}^{r}\left[\frac{\gamma}{2}-\frac{1}{2e^{-\gamma}}(\psi_i-\mu)^2\right] - \sum_{i=1}^{N-r-s}\left[\frac{\gamma}{2}-\frac{1}{2e^{-\gamma}}(\theta_i-\mu)^2\right] \\
&\qquad -\sum_{i=1}^{r}\log\left[\binom{100}{z_i}\left(\frac{e^{\psi_i+w_i}}{1+e^{\psi_i+w_i}}\right)^{z_i}\left(1-\frac{e^{\psi_i+w_i}}{1+e^{\psi_i+w_i}}\right)^{100-z_i}\right] \\
&\quad = C + \sum_{i=1}^{s}\left[\frac{e^{\gamma}}{2}(\omega_i-\mu)^2\right] + \sum_{i=1}^{r}\left[\frac{e^{\gamma}}{2}(\psi_i-\mu)^2\right] + \sum_{i=1}^{N-r-s}\left[\frac{e^{\gamma}}{2}(\theta_i-\mu)^2\right] - \frac{N\gamma}{2} \\
&\qquad -\sum_{i=1}^{r}\left[\log\binom{100}{z_i} + z_i\left[\psi_i+w_i-\log\left(1+e^{\psi_i+w_i}\right)\right] + (100-z_i)\log\left(\frac{1}{1+e^{\psi_i+w_i}}\right)\right] \\
&\quad = C + \frac{e^{\gamma}}{2}\left[\sum_{i=1}^{s}(\omega_i-\mu)^2 + \sum_{i=1}^{r}(\psi_i-\mu)^2 + \sum_{i=1}^{N-r-s}(\theta_i-\mu)^2\right] - \frac{N\gamma}{2} \\
&\qquad -\sum_{i=1}^{r}\left[\log\binom{100}{z_i} + z_i(\psi_i+w_i) - 100\log\left(1+e^{\psi_i+w_i}\right)\right]
\end{aligned}
$$

To find $U_t(\mu, \gamma, \psi)$, the formula for the negative logarithm of the posterior density for the variables μ, γ, ψ given observed data (ω, w, z) at time t, we can group terms independent of μ, γ, ψ in the joint negative logarithm density into the constant:

$$\begin{aligned} U_t(\mu, \gamma, \psi) &= -\log\left[\pi(\mu, \gamma, \psi | \omega, w, z)\right] \\ &= C + \frac{e^\gamma}{2}\left[\sum_{i=1}^{s}(\omega_i - \mu)^2 + \sum_{i=1}^{r}(\psi_i - \mu)^2\right] - \frac{(s+r)\gamma}{2} - \sum_{i=1}^{r}\left[z_i\psi_i - 100\log\left(1 + e^{\psi_i + w_i}\right)\right] \end{aligned}$$

Hence, the gradient of the above posterior becomes:

$$\nabla U_t(\mu, \gamma, \psi) = \begin{pmatrix} \dfrac{\partial U}{\partial \mu} \\ \dfrac{\partial U}{\partial \gamma} \\ \dfrac{\partial U}{\partial \psi_1} \\ \vdots \\ \dfrac{\partial U}{\partial \psi_r} \end{pmatrix}$$

$$= \begin{pmatrix} -e^\gamma\left[\displaystyle\sum_{i=1}^{s}(\omega_i - \mu) + \sum_{i=1}^{r}(\psi_i - \mu)\right] \\ \dfrac{e^\gamma}{2}\left[\displaystyle\sum_{i=1}^{s}(\omega_i - \mu)^2 + \sum_{i=1}^{r}(\psi_i - \mu)^2\right] - \dfrac{s+r}{2} \\ e^\gamma(\psi_1 - \mu) - z_1 + \dfrac{100e^{\psi_1 + w_1}}{1 + e^{\psi_1 + w_1}} \\ \vdots \\ e^\gamma(\psi_r - \mu) - z_r + \dfrac{100e^{\psi_r + w_r}}{1 + e^{\psi_r + w_r}} \end{pmatrix}$$

Question 1d) Implementing the Hamiltonean Monte Carlo Markov Chain for the above distribution and removing burn-in, the following patterns can be seen.

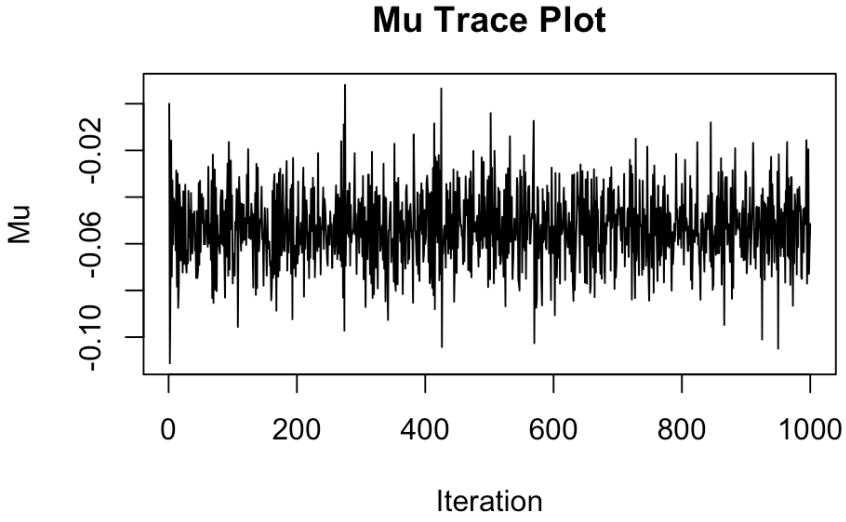


Figure 8: Trace Plot for μ

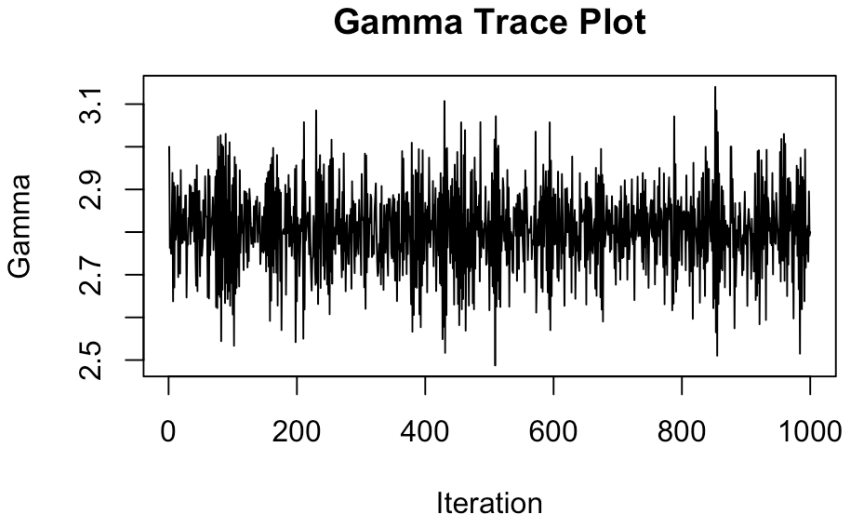


Figure 9: Trace Plot for γ

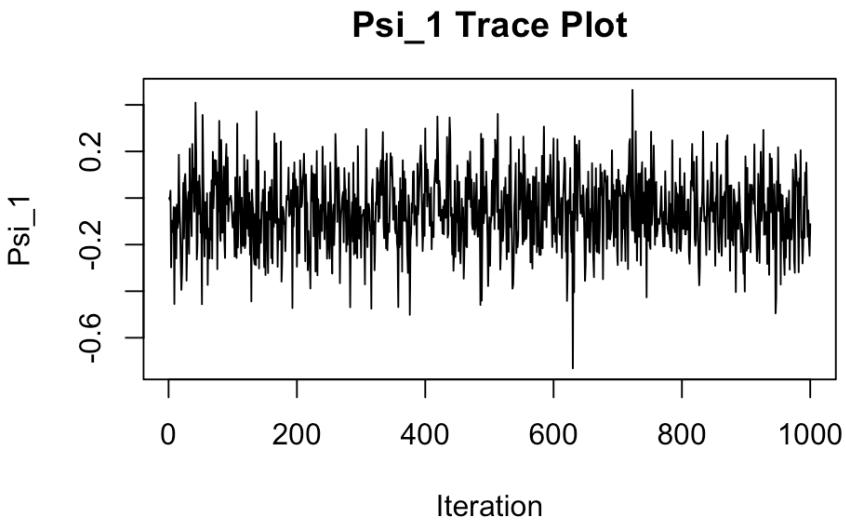


Figure 10: Trace Plot for ψ_1

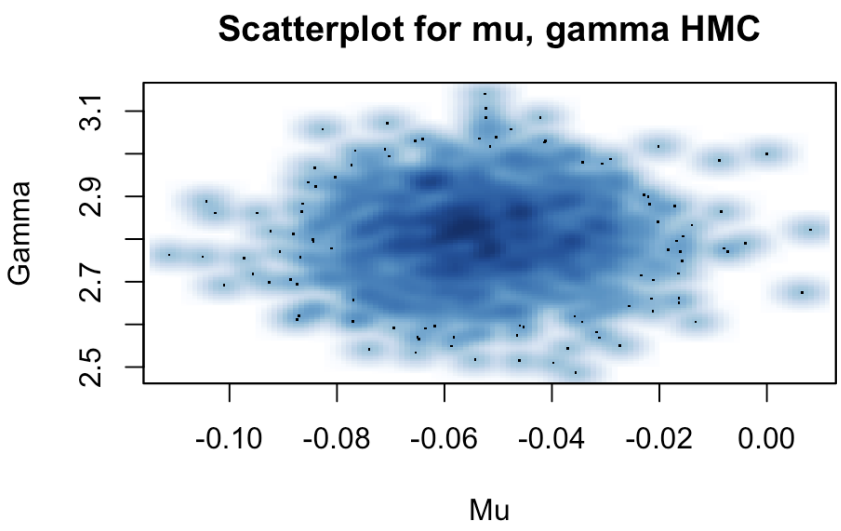


Figure 11: Scatterplot for μ vs. γ

The values of μ and γ converge around the values of -0.05 and 2.80 respectively, roughly matching the post-hoc expectations from part 1a).

Question 1e)

Since the log-odds ratio of party A winning r proportion of the vote is defined as $\log \frac{r}{1-r}$, only if the log-odds ratio is greater than 0 will the proportion r be greater than $\frac{1}{2}$ (i.e. party A will win).

Extending the function from 1d) to return the estimated proportion of times party A will win at time t given data at the time (see R code separately attached), at $t = 3.4$ given the data, it is estimated that party A will win with a probability of roughly 1.

Question 1f) Instead of looking at probability that party A is predicted to win at a single time (e.g. $t = 3.4$), by first filtering for the unique result update times, we can apply the probability function in part e) at all these times to create a plot of the probability that party A will win the election. In other words, at each time t, we look at each of the steps in the Hamiltonean Monte Carlo chain, which should have:

1. One posterior value for μ
2. One posterior value for γ

3. r posterior values for $\psi_1, ..., \psi_r$

Then, for each of these steps, we can simulate the total number of seats won by party A in said iteration

1. For the districts with final results already determined $(\omega_1, ...\omega_s)$, we directly count the number of seats won by party A
2. For the districts with exit polls but final results not yet determined $(\psi_1, ..., \psi_r)$, we use the posterior values of ψ in this iteration, add to the previous election's results w, and predict the results for these districts as $p_i = \psi_i + w_i, i = 1, ..., r$
3. For the districts with no exit polls and final results not yet determined $(\theta_1, ..., \theta_{N-r-s})$, we simulate results for each of these districts using the posterior values of μ and γ in this iteration. Then, we add this to the previous election's results q, and predict the results for these districts as $p_i = \theta_i + q_i, i = 1, ..., N - r - s$

We then add count how many districts are won by party A in each iteration above. The Monte Carlo integral will thus be the count of simulated iterations that party A won a majority, divided by the number of simulations.

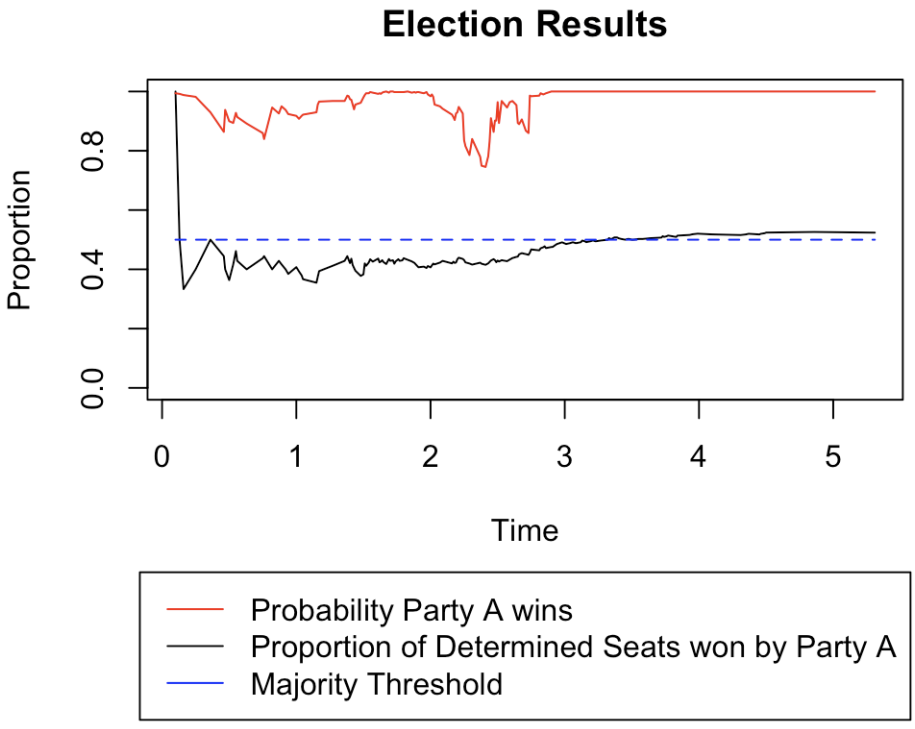


Figure 12: Election Results

Looking at the figure, the proportion of seats won by party A is less than half until $t = 3.32$, and only by looking at known results it seems like a tight election race. However, looking

at the probability party A is predicted to win, surprisingly this probability never falls below the 50% prediction rate - the worst prediction of party A winning occurs around 2.5 hours in, and is still above 70%. Of course the credibility of this prediction has is based on many assumptions (e.g. in our model, choices made for the HMC parameters, etc.), but nonetheless this was interesting to see.